

Landau–Zener–Weber Scattering: Exact Connection Data and Finite-Window Unitary Controls

HCS Research Program

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Abstract

We analyze the genuinely nonautonomous two-level crossing $i\dot{\psi} = [(vt/2)\sigma_z + g\sigma_x]\psi$, $v > 0$. Eliminating one component gives a pair of parabolic-cylinder (Weber) equations. Their connection formula yields, in a fixed diabatic gauge, the exact asymptotic survival law $P_{\text{diab}} = \exp(-2\pi g^2/v)$ and the Stokes/Gamma phase $\phi_S = \pi/4 + \delta(\log \delta - 1) + \arg \Gamma(1 - i\delta)$. Wronskian conservation proves an $SU(2)$ scattering matrix, while elementary calculus gives strict monotonicity and sudden/adiabatic limits. A separate 80-digit, fixed-step RK4 ledger checks finite windows without calling them an exact finite-time formula. The source-local theorem is a unitary/scattering candidate only: it supplies no arithmetic owner, target divisor, or Hilbert–Polya bridge.

1 Crossing and conventions

Set $\hbar = 1$, $\sigma_z = \text{diag}(1, -1)$, and use the diabatic basis at both ends of time:

$$H(t) = \frac{vt}{2}\sigma_z + g\sigma_x, \quad i\frac{d}{dt}\begin{pmatrix} a \\ b \end{pmatrix} = H(t)\begin{pmatrix} a \\ b \end{pmatrix}, \quad v > 0, \quad g \in \mathbb{R}. \quad (1)$$

The physical clock is t ; there is no fitted repetition or target clock. We write $\delta = g^2/v$ and fix the asymptotic phase convention by the matrix in Theorem 1. Conjugating by σ_z maps g to $-g$.

Theorem 1 (Weber reduction and scattering law). *For $g \neq 0$, the components of a solution of (1) satisfy*

$$a'' + \left(g^2 + \frac{v^2 t^2}{4} + \frac{iv}{2}\right)a = 0, \quad (2)$$

$$b'' + \left(g^2 + \frac{v^2 t^2}{4} - \frac{iv}{2}\right)b = 0. \quad (3)$$

After the usual rotation of t and affine rescaling these are Weber parabolic-cylinder equations. Unit-flux connection data from $t = -\infty$ to $t = +\infty$ are

$$S(\delta) = \begin{pmatrix} \sqrt{P} & -\sqrt{1-P}e^{i\phi_S} \\ \sqrt{1-P}e^{-i\phi_S} & \sqrt{P} \end{pmatrix}, \quad P = e^{-2\pi\delta}, \quad \phi_S = \frac{\pi}{4} + \delta(\log \delta - 1) + \arg \Gamma(1 - i\delta), \quad (4)$$

with $\phi_S(0) = \pi/4$ by continuity. A different diagonal asymptotic phase convention conjugates S by diagonal unitaries and leaves P intact.

Proof. From the first component equation, $b = (ia' - (vt/2)a)/g$. Substitution into the second gives the first equation in (3); exchanging components gives the second. The rotated equation is the standard parabolic-cylinder equation. Its connection identity and constant Wronskian produce the two entries in (4); the modulus identity $|\Gamma(1 + i\delta)|^2 = \pi\delta/\sinh(\pi\delta)$ reduces the diagonal coefficient to $e^{-\pi\delta}$. This is a source-local special function calculation, with no external spectral data. $\square$

Proposition 1 (Unitary and monotone invariants). *For real parameters, $H(t) = H(t)^*$ and every finite-time propagator preserves the Hermitian norm. The Weber Wronskian gives the same flux identity at the two ends, so $S^*S = I$ and $\det S = 1$. Moreover*

$$\frac{dP}{d\delta} = -2\pi e^{-2\pi\delta} < 0 \quad (\delta > 0), \quad \left.\frac{\partial P}{\partial g}\right|_v = -\frac{4\pi g}{v}e^{-2\pi g^2/v}, \quad (5)$$

and $\phi'_S(\delta) = \log \delta - \Re\psi(1 - i\delta)$.

Proof. Hermiticity gives $(\psi^*\psi)' = 0$; the Wronskian is the equivalent constant flux in the Weber basis. Direct multiplication of (4) proves the $SU(2)$ identities. Differentiating $e^{-2\pi\delta}$ gives (5); differentiating $\log \Gamma(1 - i\delta)$ gives the phase derivative. $\square$

2 Limits, boundaries, and a finite-window control

The crossing parameter organizes the singular faces. As $\delta \downarrow 0$,

$$P = 1 - 2\pi\delta + O(\delta^2), \quad 1 - P = 2\pi\delta + O(\delta^2), \quad (6)$$

so the sudden limit $v \rightarrow \infty$ is diabatic. At fixed $g \neq 0$, $v \downarrow 0$ gives $P \rightarrow 0$ (adiabatic following). At $g = 0$ the two channels decouple exactly, and the continuous phase convention is used. The sign change $g \mapsto -g$ is a constant σ_z gauge, not a new probability law. The point $t = 0$ is the sole diabatic gap closing; it is the turning point of the Weber connection problem.

For validation, we integrated both basis vectors of (1) on $[-T, T]$ with a fixed 2048-step classical RK4 scheme at 80 decimal digits. The ledger records all four complex entries, $P_{\text{window}} = |U_{11}|^2$, $|P_{\text{window}} - P|$, and $\|U^*U - I\|_{\max}$. It is a controlled numerical window, not an exact finite-time propagator. Representative exact rows are:

case	v	g	δ	P_{diab}	ϕ_S
reference	1	1/2	1/4	0.2078795764	0.3270619133
fast/weak	4	1/3	1/36	0.8398492016	0.6741033767
slow/strong	1/4	3/4	9/4	0.0000007249	0.0372977224
negative g	2	-1/2	1/8	0.4559381278	0.4718436007
uncoupled	3/2	0	0	1	0.7853981634

Table 1: Exact source-local scattering rows in the fixed gauge.

The finite ledger has 15 rows ($T = 2, 4, 8$ for each case). The largest Gram residual is 1.56×10^{-5} with the 2048-step control; discrepancies are reported as finite-window effects. A producer-independent checker recomputes the ODE, while a separate SymPy program checks the signs in (3), the Pauli square, $SU(2)$ determinant, monotonicity, and the coupling gauge. The machine-readable field is written literally as `P_diabatic`; the scope lock is `NO_BAD_EULER_OR_ROOT_NUMBER`.

3 Relation to the Route-A decision

The theorem is a complete physical scattering result, but a single swept crossing has no primitive periodic-orbit repetition law. The parameter δ is a source coupling ratio, not an arithmetic label; P and ϕ_S are not target zeros or divisor data. Thus the strict tuple is

$$(\text{A0_FAIL}, \text{A1_FAIL}, \text{A2_FAIL}, \text{A3_FAIL}, \text{A4_UNITARY_OR_SCATTERING_CANDIDATE}),$$

with `overall=ROUTE_A_REJECTED` and `route_b_invocation_allowed=false`. This nonautonomous model is intentionally distinct from the autonomous Jaynes–Cummings excitation blocks of C223. No target operator, Euler factor, root number, automorphy claim, or Hilbert–Polya bridge is made.

Reproducibility and source note

The receipt is generated by `code/c224_landau_zener_producer.py` using exact rational sentinels, 80-digit Gamma evaluation, and deterministic RK4. The independent checker, symbolic cross-check, clean replay, and hostile mutation suite are run in separate processes. The fixed-epoch LuaLaTeX build is repeated twice; the three revision PDFs are content-distinct and the final `main.pdf` is byte-identical to round 2. The release manifest closes exactly 27 payload files and excludes itself and build sidecars.

The source audit records the following DOI-verified references:

References

- [1] C. Zener, “Non-adiabatic crossing of energy levels,” *Proceedings of the Royal Society A* 137 (1932), 696–702. DOI: 10.1098/rspa.1932.0165.
- [2] N. V. Vitanov and B. M. Garraway, “Landau-Zener model: Effects of finite coupling duration,” *Physical Review A* 53 (1996), 4288–4304. DOI: 10.1103/PhysRevA.53.4288.
- [3] S. N. Shevchenko, S. Ashhab, and F. Nori, “Landau-Zener-Stuckelberg interferometry,” *Physics Reports* 492 (2010), 1–30. DOI: 10.1016/j.physrep.2010.03.002.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic or Hilbert–Polya claim. **Data and code.** All rows and audits are released with HCS-C224. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked displayed claims and metadata. This is not external peer review.